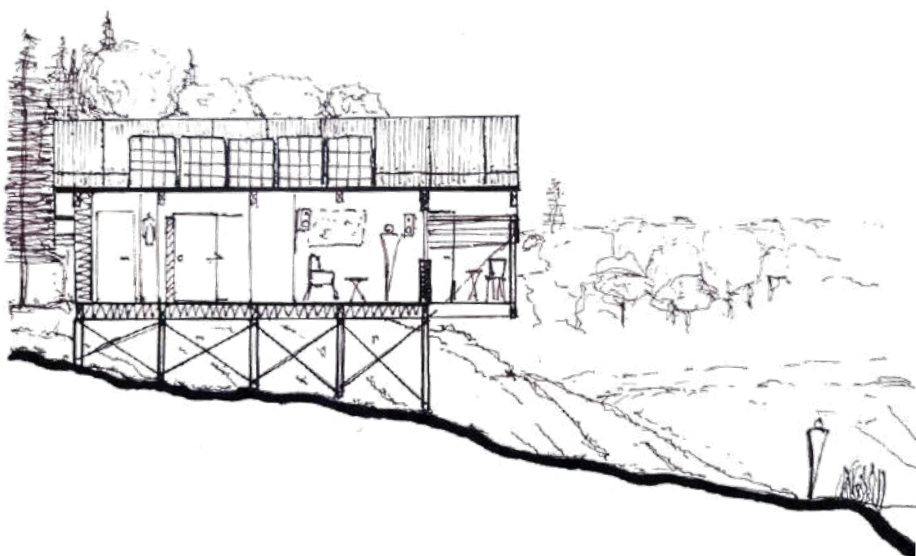




*Academic
works.*

Contents

Museum Jorn page 1

A large Hybrid building.

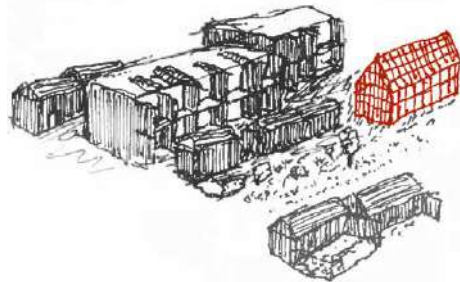
- Daylight-optimiszd parametric roof



Drivhusene..... page 7

A sustainable settlement

- Co-living
- Be18
- Energy optimization

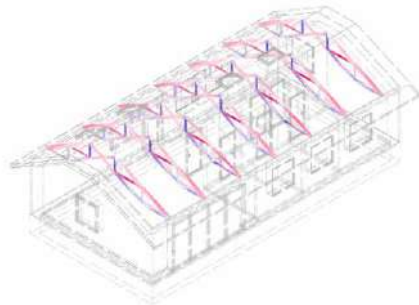


Designing with Scarcity..... page 13

Algorithmic Workflows for Structural
Optimization Using Reclaimed Timber

[- Github](#)

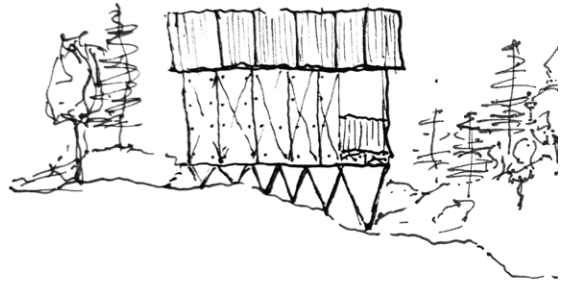
[- GIF](#)



Writers Retreat..... page 19

A self sufficient cabin.

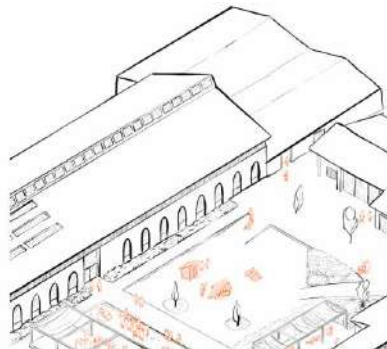
- Energy optimization, shutters, openings and solar panels



Tranformation..... page 25

School and community hub

- Energy optimization



Extra..... page 29

Daylight, DGNB

LCA

Robotic fabrication

Clayprint

Museum Jorn

A large Hybrid building.



**Bachelor's Project, Architecture and design, Aalborg University,
Designing a Large Hybrid Building, S23
Group Project**

The modern urban landscape demands a diverse array of functions within its large buildings to effectively cater to the needs of its populace on a social scale. The concept of hybrid building typology aims to address these demands.

Situated in Silkeborg, Denmark, the project endeavors to create a new Museum Jorn that embodies the principles of hybrid building, drawing inspiration from the proposed plans for Søfronten by Arkitektura and Silkeborg Kommune.

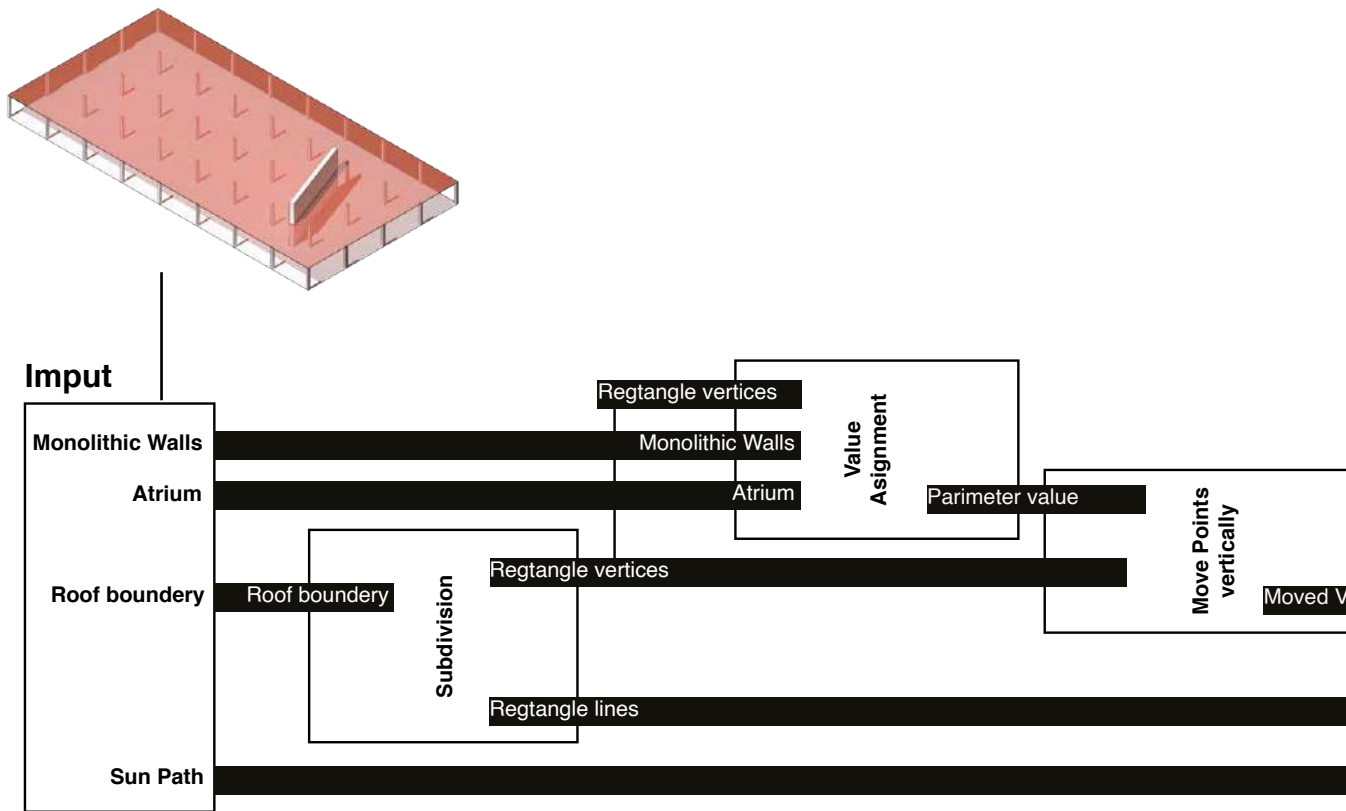
To achieve these objectives on an urban scale, the design integrates the city center with the planned urban development on Søfronten, utilizing the building's volume to harmonize with Silkeborg's topology and road network, thus bridging height differences.

The architectural concept, influenced by Jorn's experimental art philosophy, challenges the conventional narrative of museum typology. This is achieved through a dynamic arrangement of walls within a static column grid, reinforced by monoliths to ensure structural stability.

The envisioned modernized Museum JORN serves as a solution for a hybrid building that blurs the boundary between exterior and interior spaces, while embodying the experimental spirit of Asger Jorn. Positioned as a visual spectacle on Søfronten, JORN symbolizes the fusion of art, architecture, and the city.



Digital maquet model

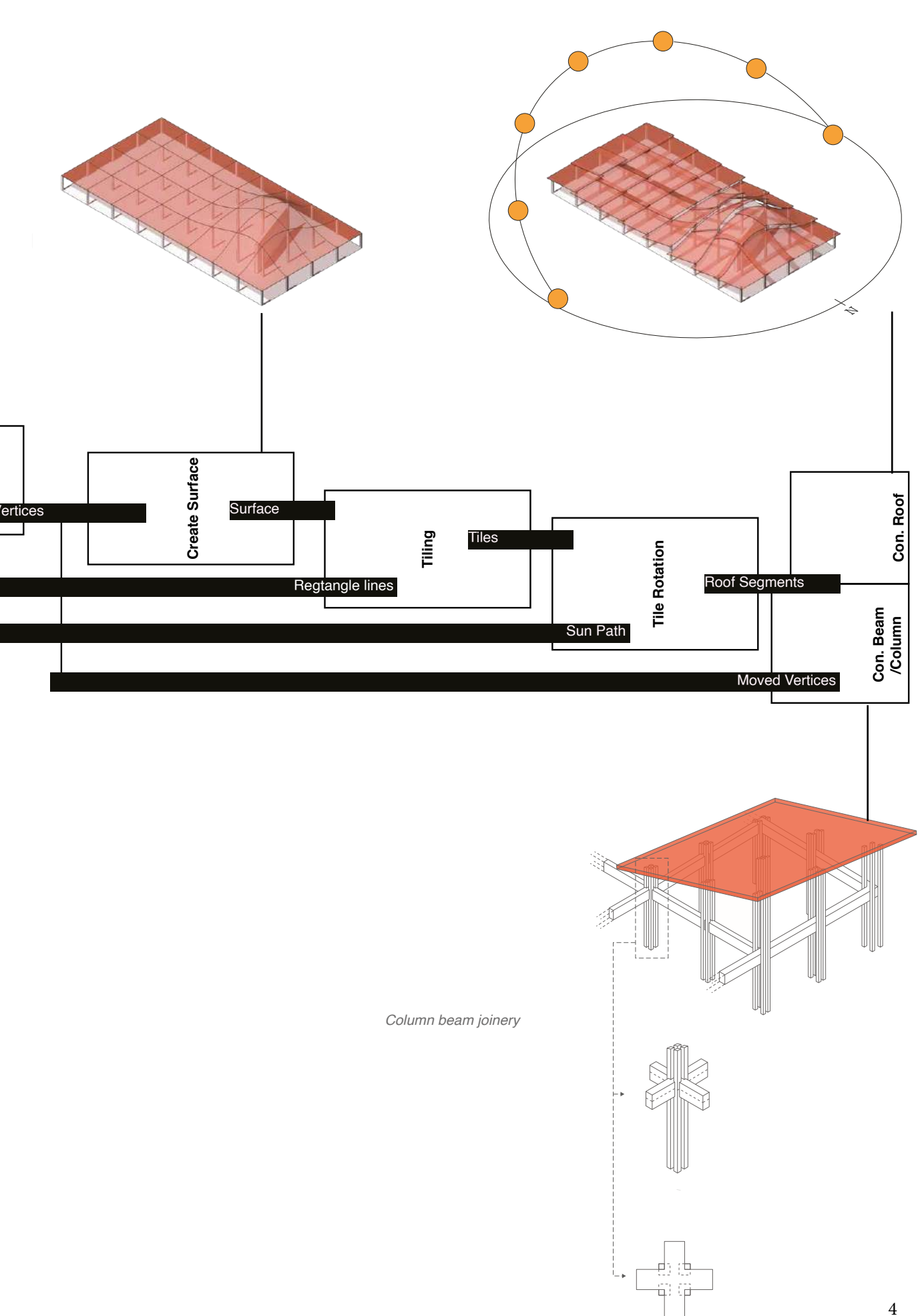


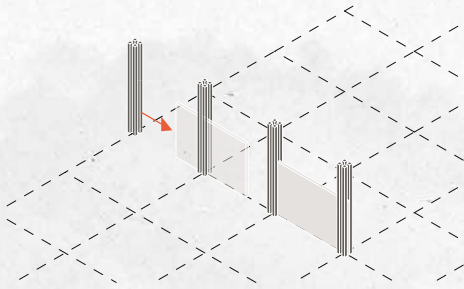
Parametric design with Rhino/Grasshopper

In the building, it is desired to exhibit a stage curtain on each floor. The stage curtains have a height of 10m, while the building's floors are 5m. The monoliths create atriums around themselves, on the ground floor and 1st floor respectively, facilitating the exhibition of the stage curtains. On the 2nd floor, the roof is designed so that the room's height is adjusted around the monolith, facilitated by the columns.

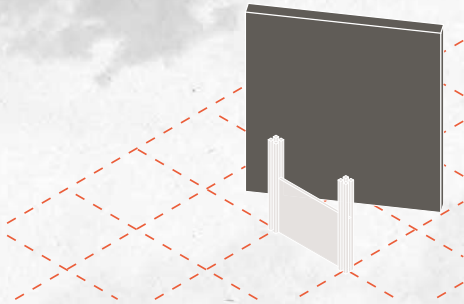
The roof is formed by the continuous columns in the building, which are extended where an increase in room height is desired.

This divides the roof surface into segments. These open towards the north, with the purpose of utilizing daylight in the building and achieving the exhaust of hot air for displacement ventilation. To promote the functionality of the columns, it is decided to further divide the roof segments. This follows the grid formed by the columns. This causes greater flexibility in the design of the roof, as well as a greater possibility of adapting the skylights as desired.

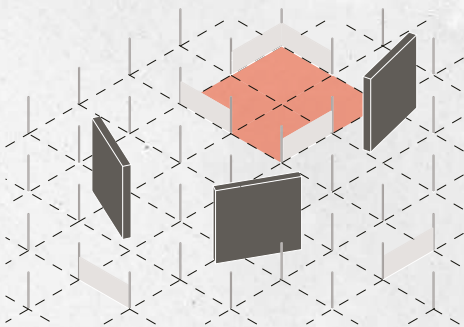




Moveable walls facilitated by columns

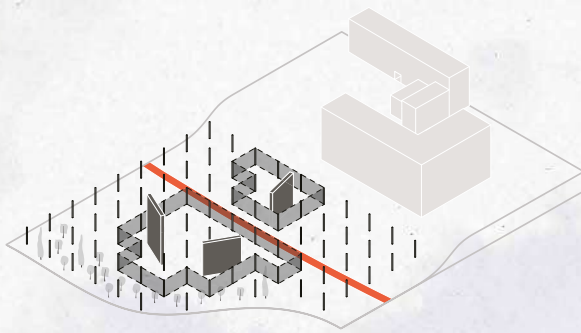


Grid contrasts the monolithic walls

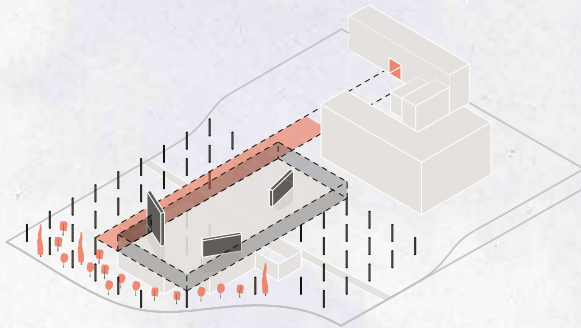


System repeats and space appears

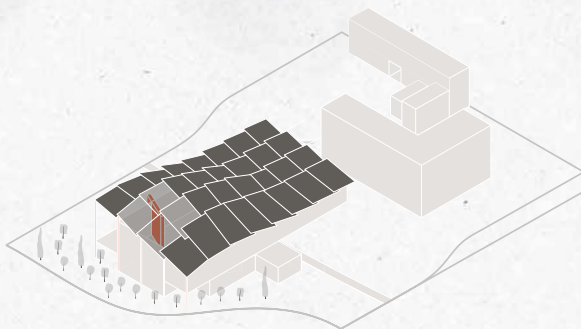




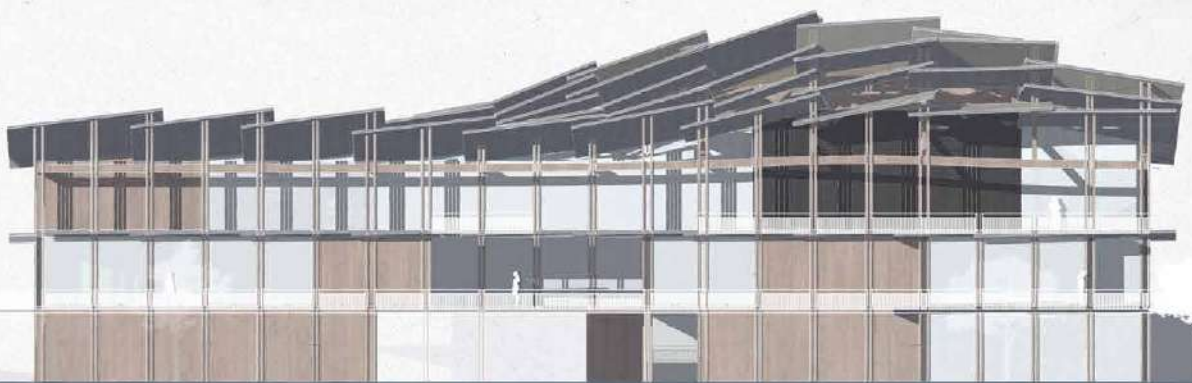
The buildings form is created by the column system



Connection to the city take shape by the column system

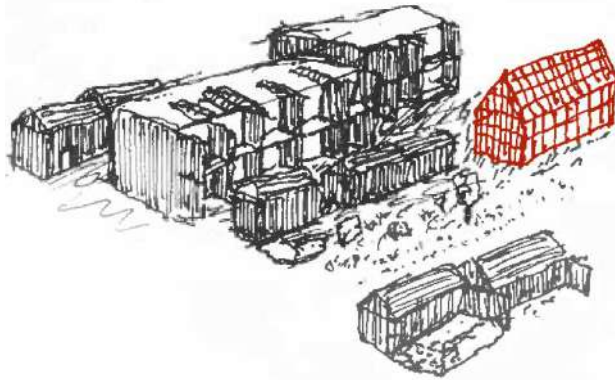


Roof is formed over the monolithic walls



Drivhusene

An sustainable settlement.



Architecture and design, Aalborg University,
Co-Living, S23
Group Project

The aim of the project is to develop a **sustainable residential settlement** on a roughly two-hectare plot located in Sorthøj, south of Aalborg, referred to as “Drivhusene.”

This co-housing settlement, at a BP of 75%, seeks fosters a sense of community among participants through low-density typologies and the integration of three common houses with varied functions.

The settlement follows a 35-degree southwest tilted grid, providing a clear structure despite various displacements and height variations.

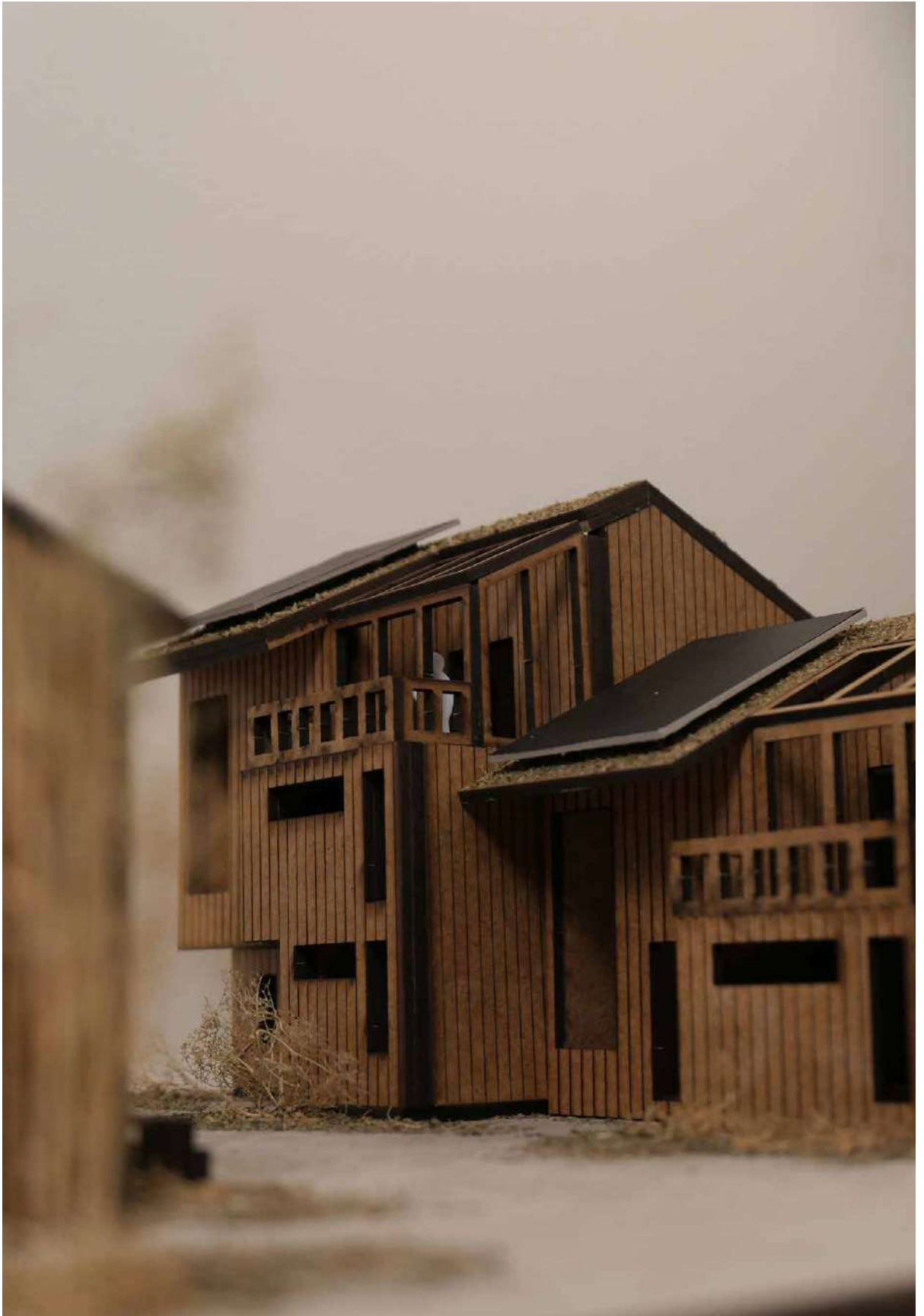
Facade orientation maximizes evening sun exposure while generating sufficient electricity through roof-mounted solar panels.

Rowhouses, the primary typology, comprise four to six units resembling a traditional Scandinavian village, with burnt wood for social outdoor spaces

and light wooden lamellas elsewhere.

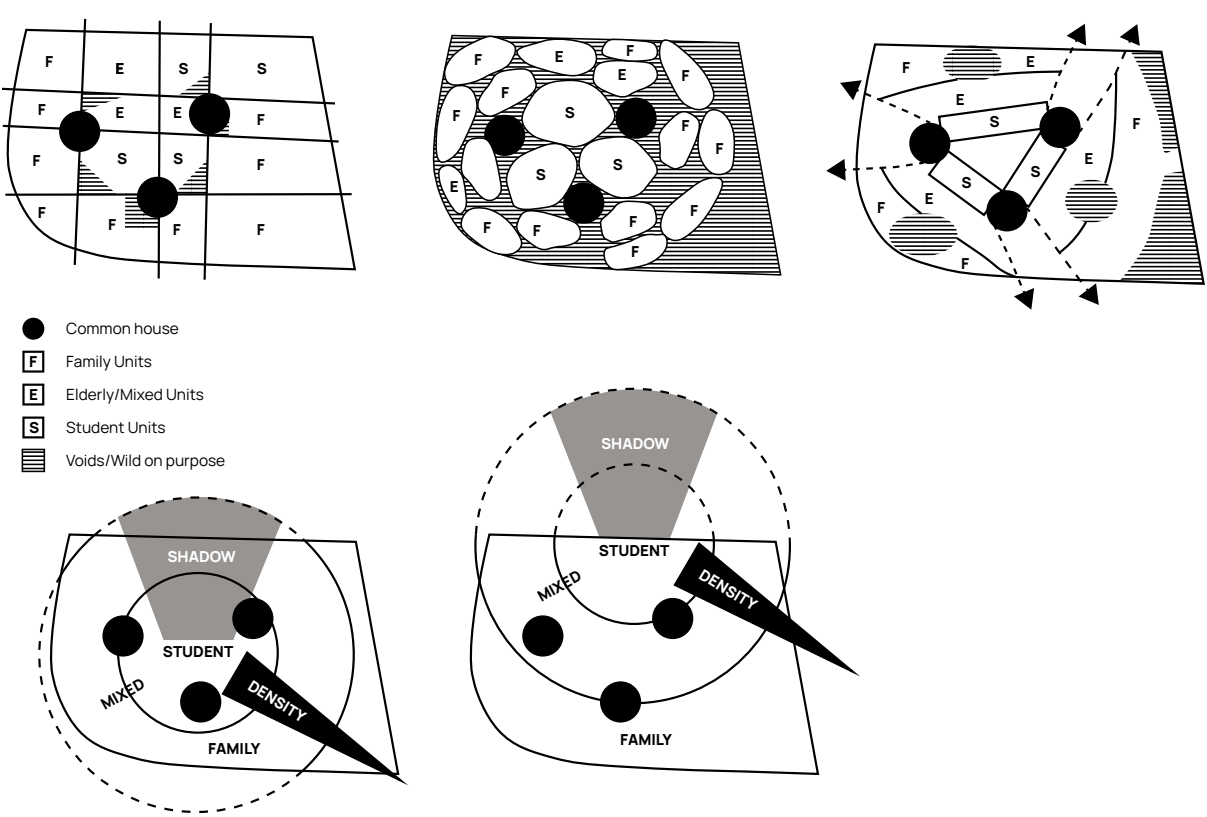
Three common houses stand out among low-density units, fostering recognition and community.

The settlement conforms to the mild topographical slope, with multiple access paths despite the main entry being to the east. CLT slabs form the structural system of rowhouses, with solar panels following the moss-covered roof shape.

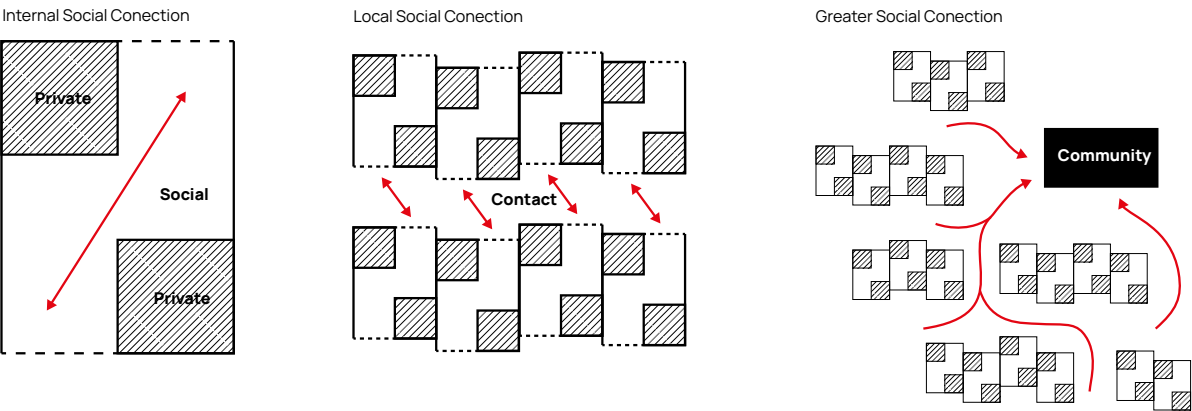


Rowhouse, Model

With a focus on social well-being, the zoning was designed to prioritize the interconnection and roles of the users within the settlement. The placement of the three common houses, voids/Wild on Purpose, and the density were also carefully considered.

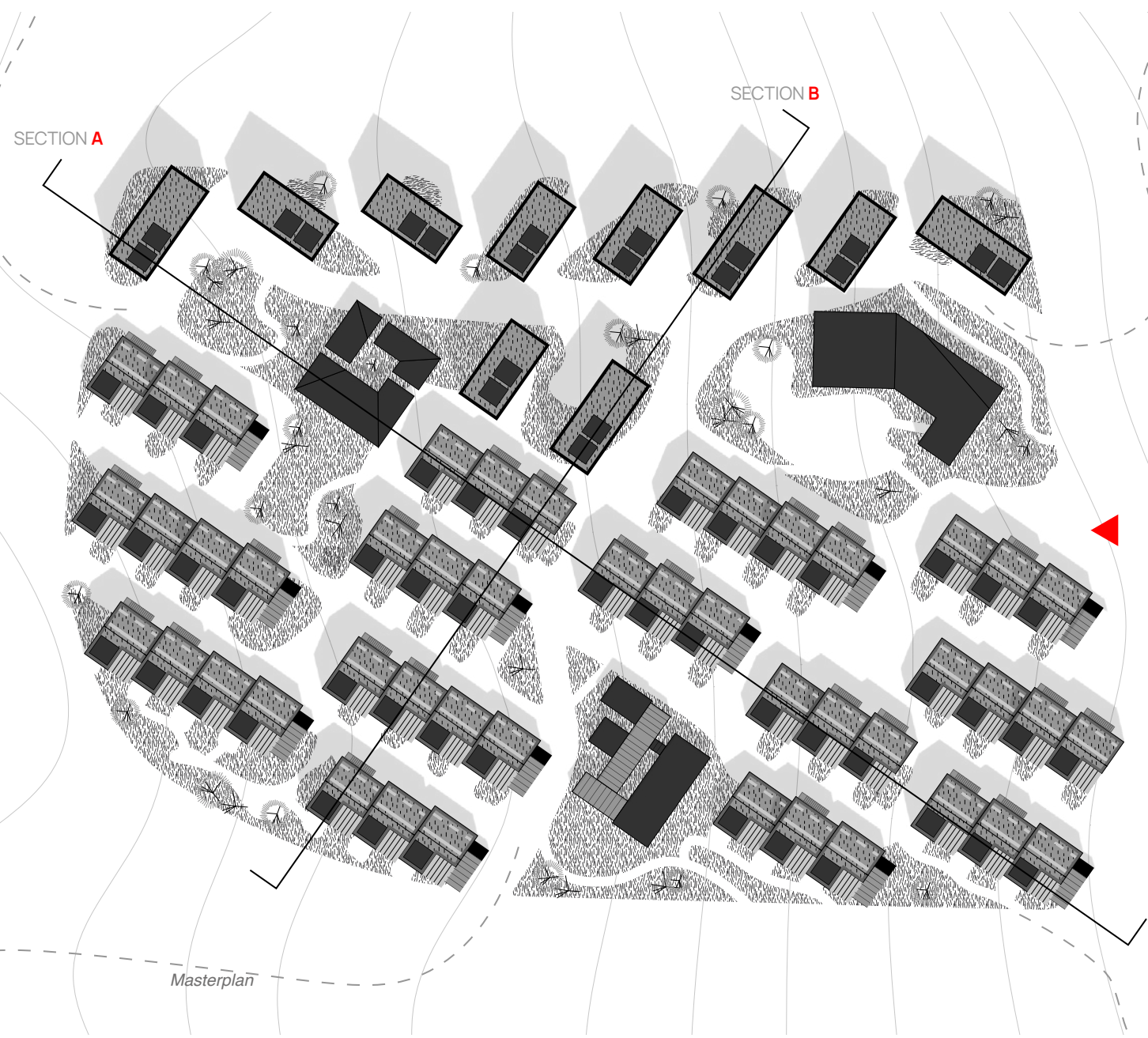


Rowhouses connectivity on multiple scales, both visually and physically.

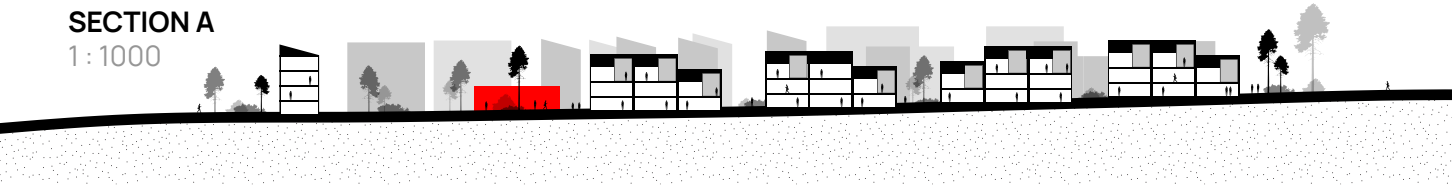


SECTION A

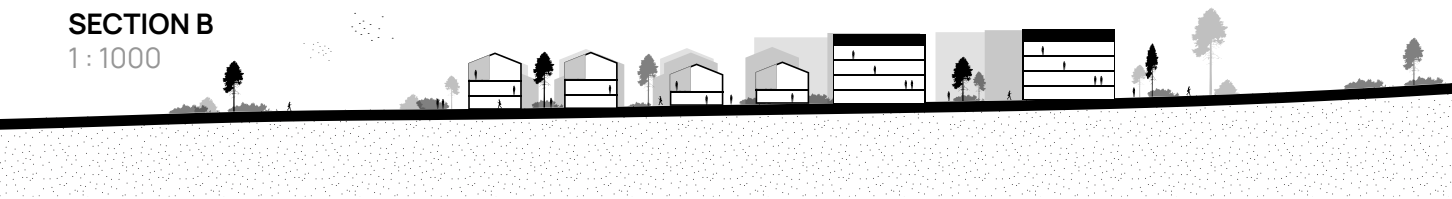
SECTION B

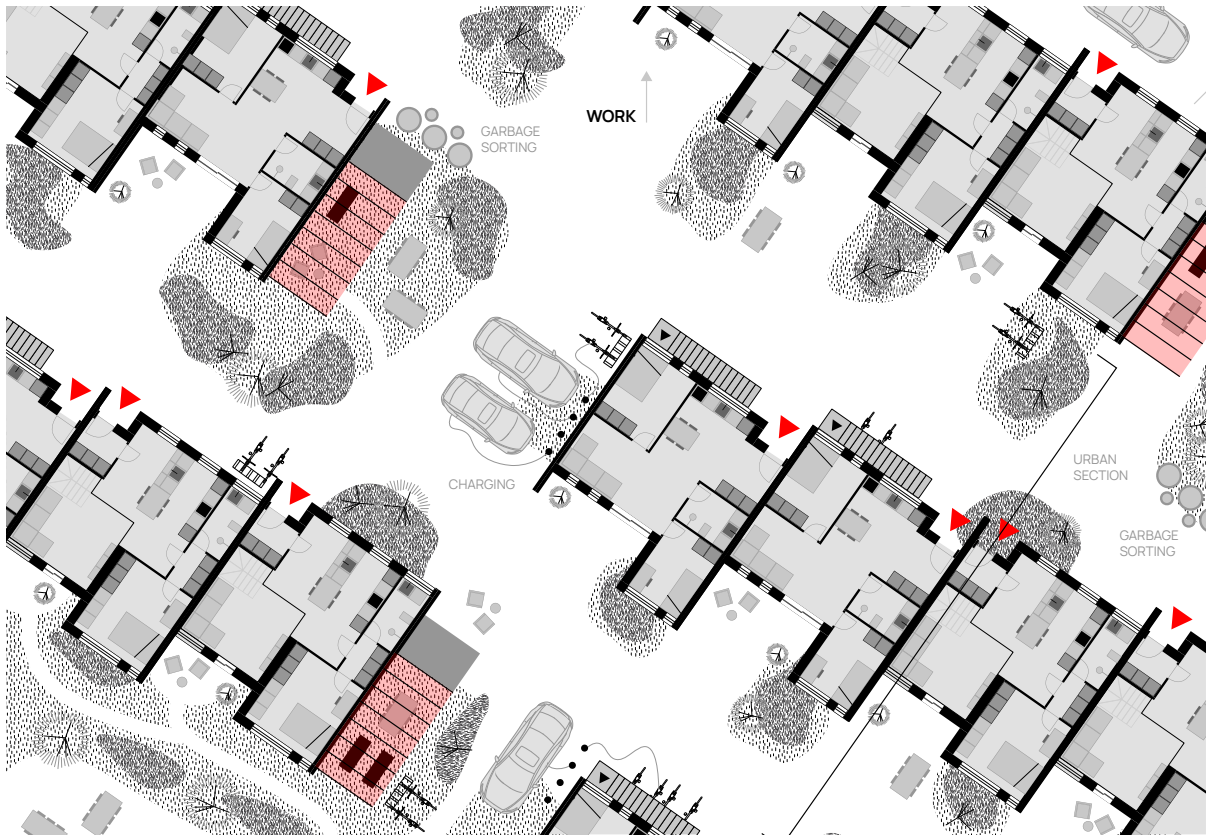


SECTION A
1:1000



SECTION B
1:1000





Sommer intersection, Render



Designing with Scarcity

Algorithmic Workflows for Structural Optimization Using Reclaimed Timber

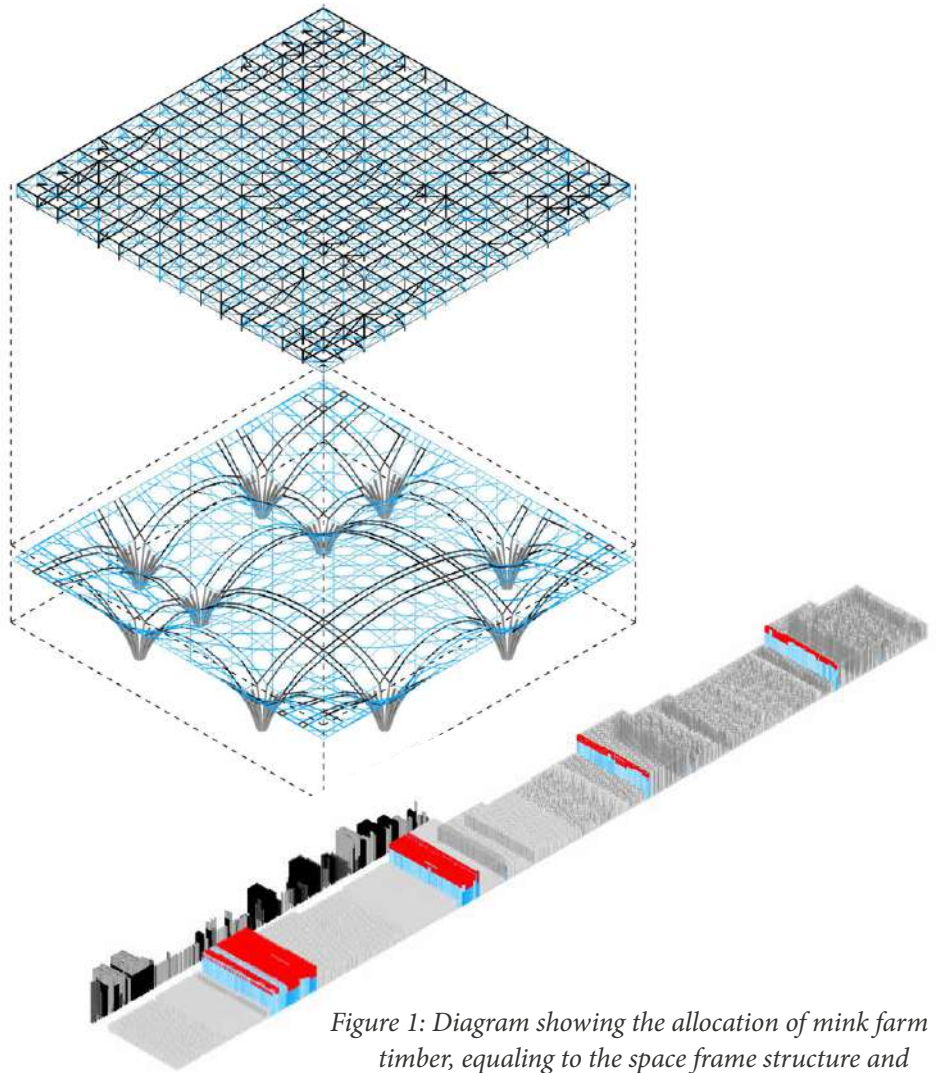


Figure 1: Diagram showing the allocation of mink farm timber, equaling to the space frame structure and the stock usage.

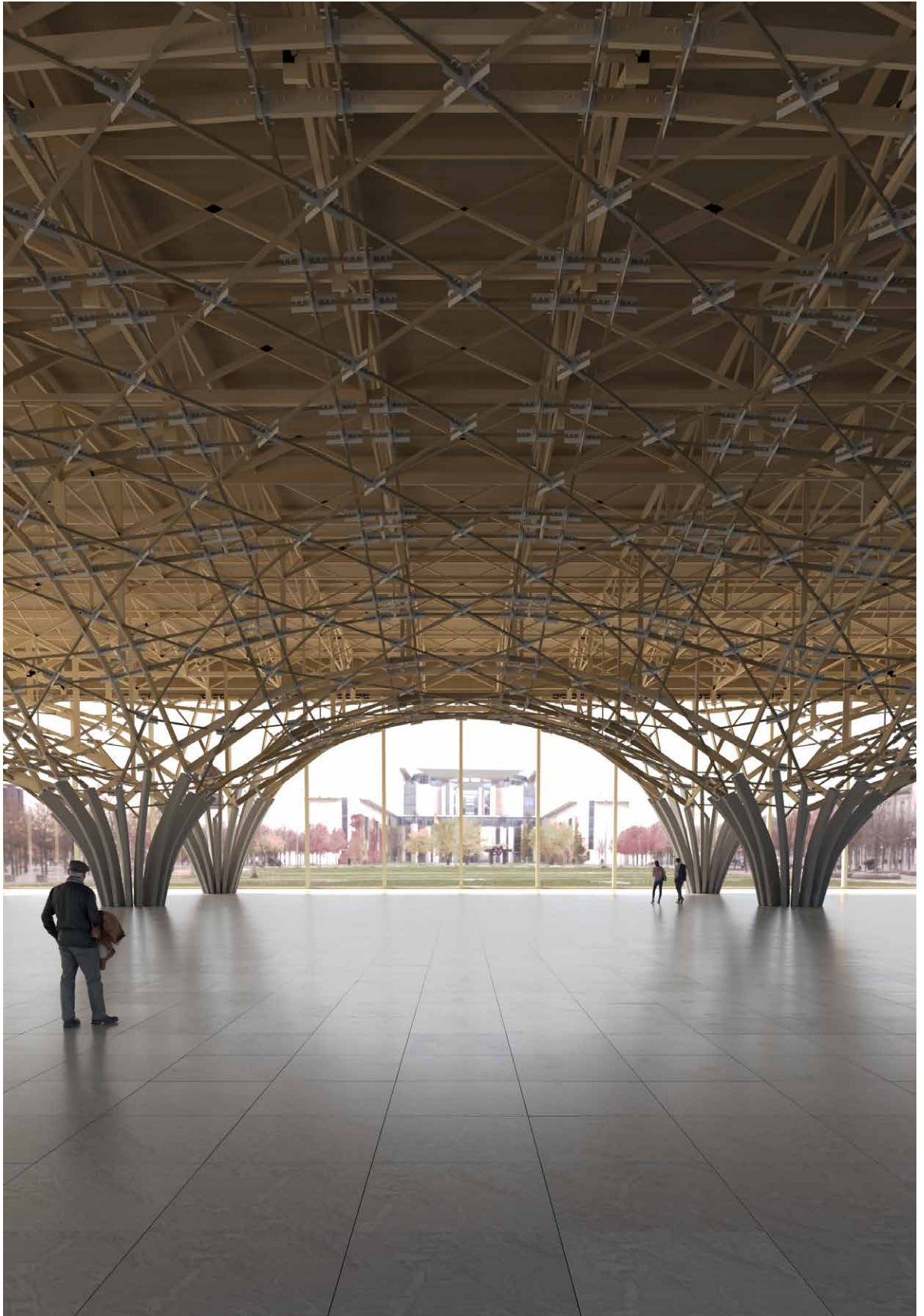


Figure 2: Interior rendering of structure #1

Method used on simple truss

This design study explores how the tool performs within a typical architectural workflow, where spatial intent, structural demands, and material constraints intersect. The case explores how the tool supports early design decisions without overriding architectural control.

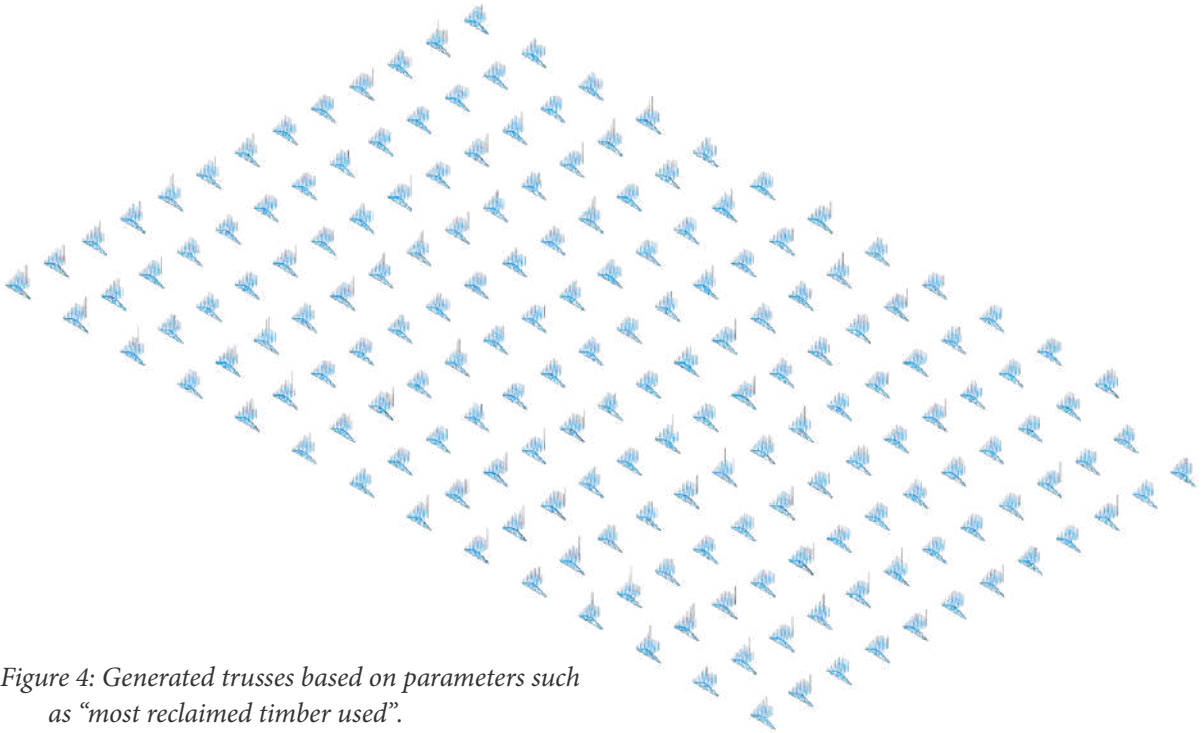


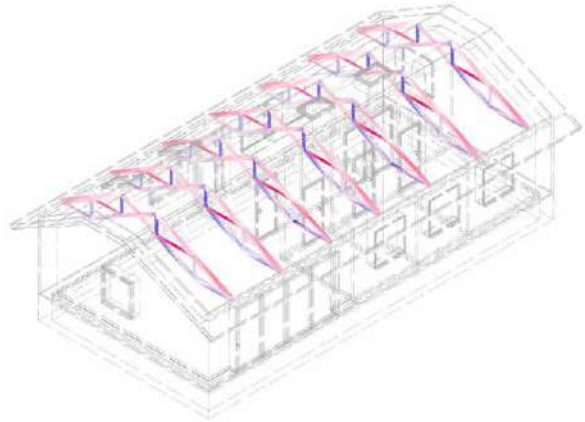
Figure 4: Generated trusses based on parameters such as “most reclaimed timber used”.



Figure 3: Top 30 choices

Tool Integration for Cross Section Optimization

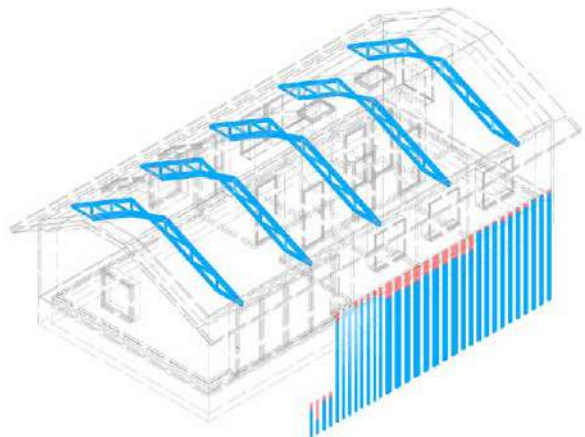
To meet fire safety regulations (30-minute integrity), the reclaimed timber elements were assembled using the Brettstapel (DLT) method. This increased cross-sectional mass, compensating for the lower performance of aged timber and enabling its use in exposed structural applications. The tool was used to calculate the needed cross sections to prepare for matching with the stock.



Tool Integration for Timber Matching & Optimization

Continuing the tool-based workflow, structural members were matched with the available reclaimed stock, allocating necessary cross-sections sourced from the mink farm inventory. Initial results showed a 75% reuse rate.

To improve this, parameters like truss density, spacing, and ridge width were adjusted and evaluated with the tool. Hundreds of structural variants were generated using multi-objective optimization, balancing the goals of minimizing both new timber use and overall weight. Two optimal outcomes were identified: one minimizing structural weight, and the other achieving 100% reclaimed timber use.



Result

The structure with 100% reclaimed elements was selected as the final design, aligning with the thesis' sustainability objectives, despite spatial compromises.

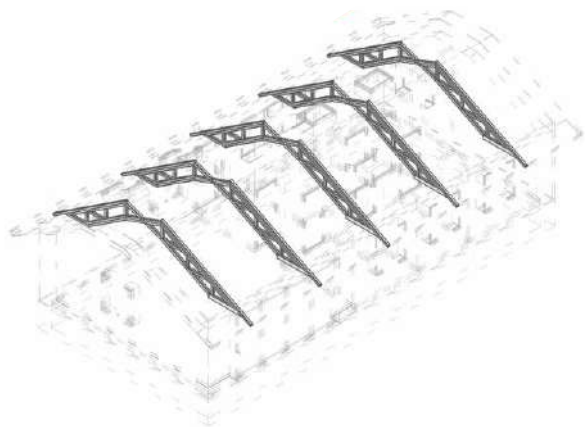


Figure 5: Diagram showing a summary of the design process.

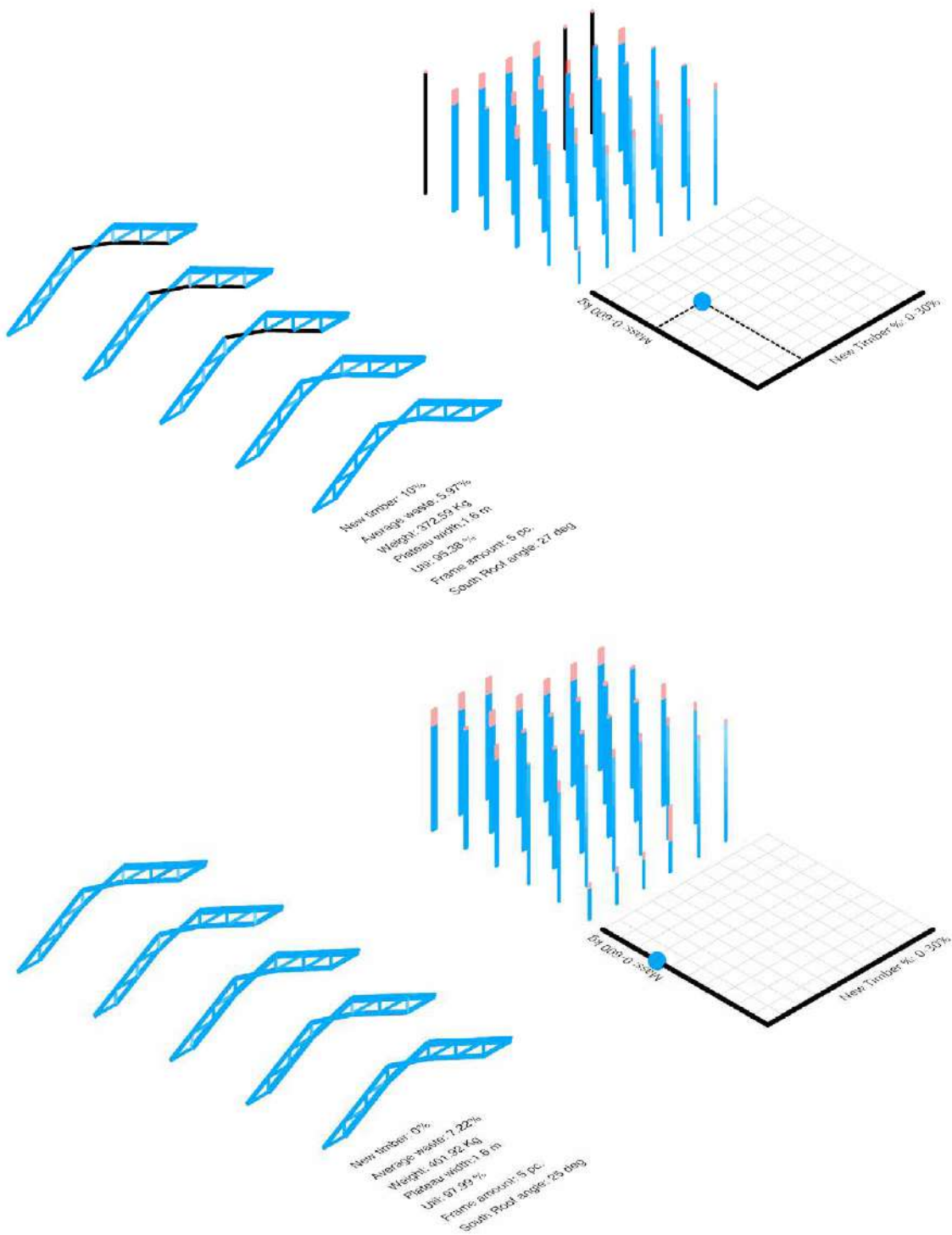


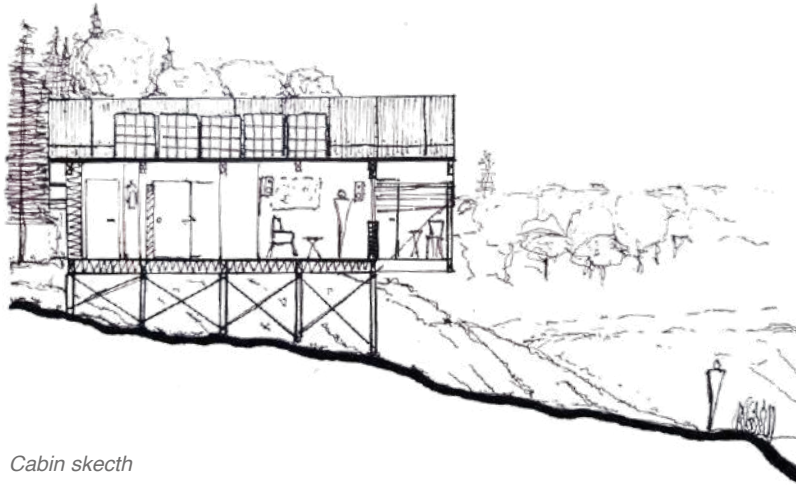
Figure 6: Diagram of the two best performing structures, iteration A (above) and iteration B (below), showing their utilization of the stock and notable data associated with the iterations, particularly the percentage of new timber, average cut-off waste and weight.



Figure 7: Visualizations of the initial structure (above) and the optimized structure (below).

Writers Retreat

A selfsufficient cabin.



Cabin skecth

MSC01 ARC - Architecture and design, Aalborg University, Advanced Integrated Design I:
Sustainable-tectonic Design
with Integrating of Structure, Site and Climate Conditions - S23
Solo Project

The retreat, **designed as a sustainable haven for writers**, offers a compact cabin with a spacious work area and terrace, powered by photovoltaics for year-round comfort.

It **prioritizes minimal energy consumption**.

Designed to engage the senses and integrate with the landscape, it embodies sustainability and functionality, offering a serene retreat for creative inspiration.

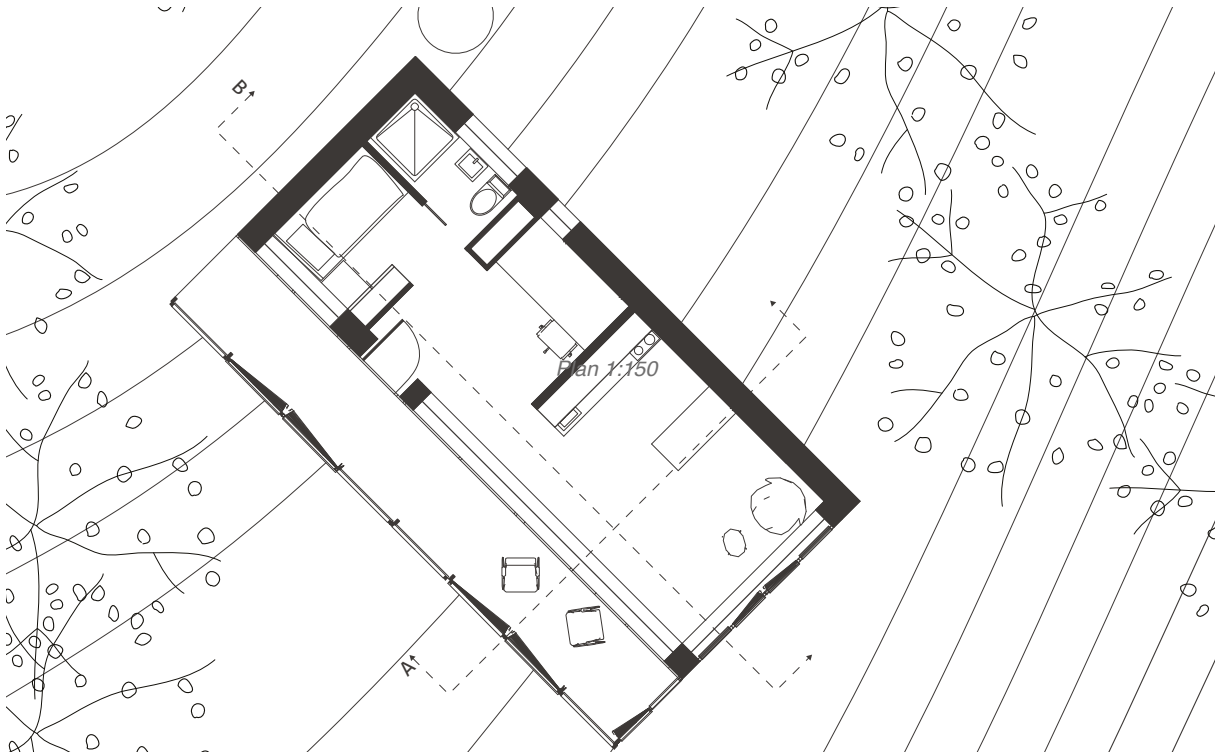
The cabin, resembling a small traditional longhouse, is elevated to interact with the forest surroundings, overlooking a pond with floating islands. Its minimal footprint on nature is emphasized by cantilevering over the hill, blending seamlessly among the trees. Guests approach on foot through vegetation, greeted by closed façades that open to the south, inviting light and controlled heat gain by moving the tactile shutters.



Cabin, Render



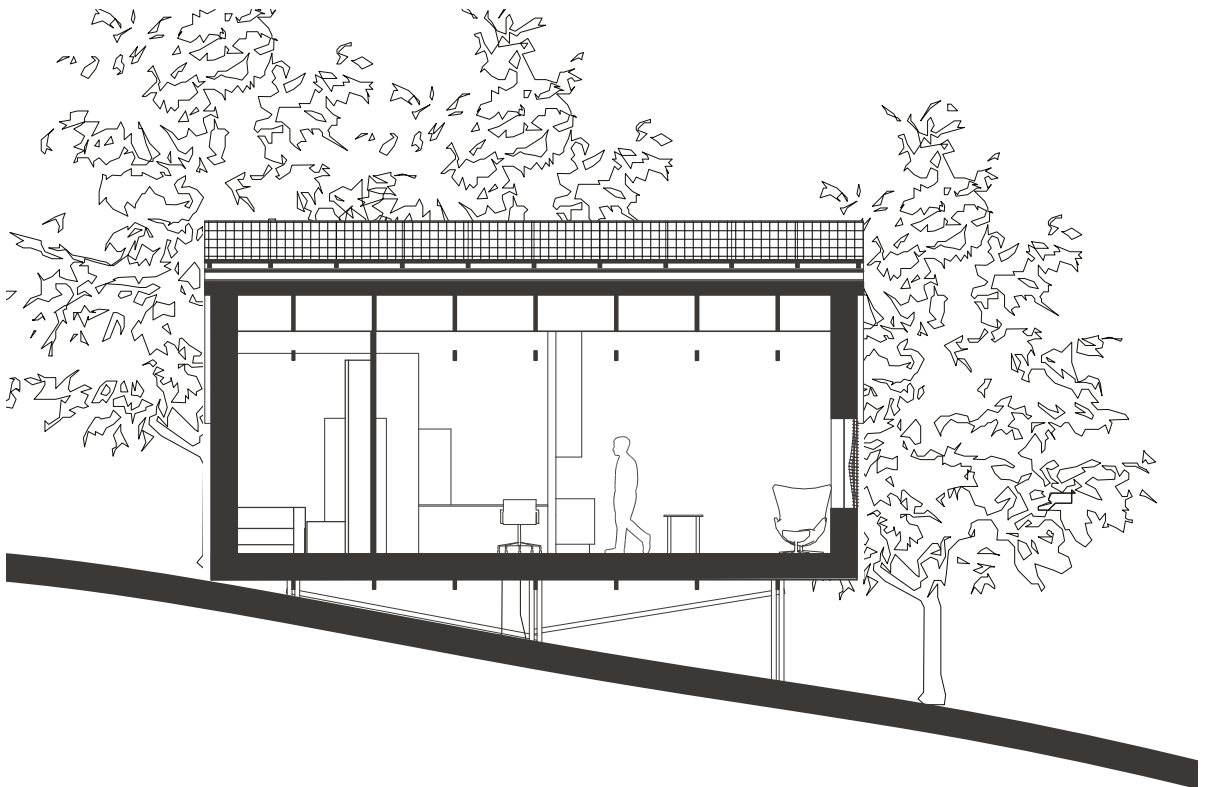
Situationplan



Plan 1:150



Section A 1:150



Section B 1:150

Roof

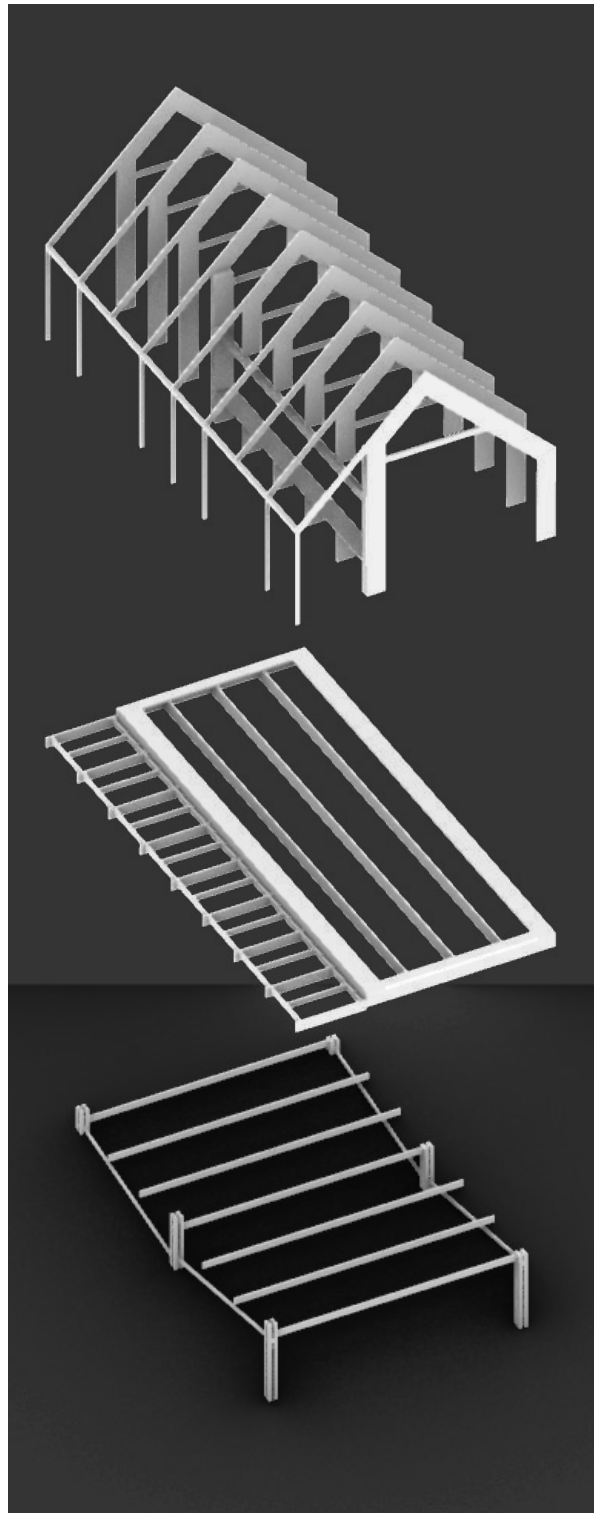
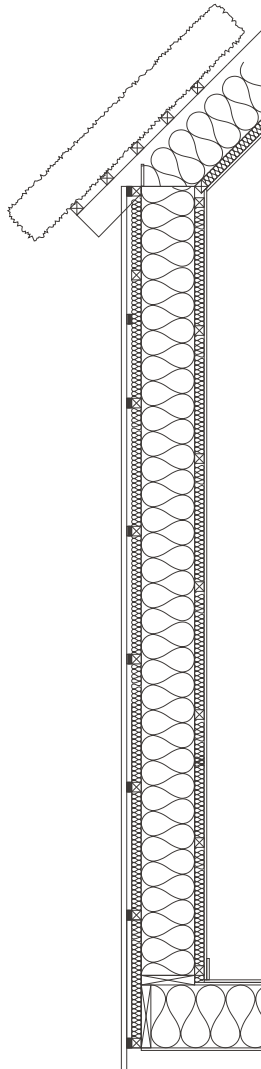
1. Ealgrass
2. Counter batten 45mm
3. Roof underlay
4. Ventilation Cavity
5. Rafter 295mm +
Insolation 245mm
6. Vaper Barrier
7. Insolation +
counter batten 45mm
8. Board cladding pine

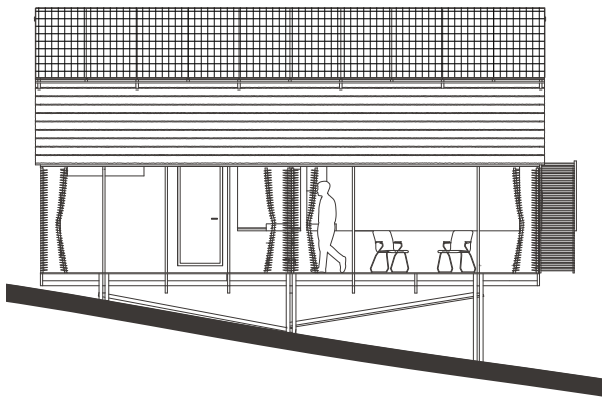
Wall

1. Facade boards
2. Ventilation batten 23mm
3. Wind barrier
4. Insolation +
Counter batten 45mm
5. Insolation +
Wood studs 245mm
6. Vaper Barrier
7. Insolation +
Counter batten 45mm
8. Board cladding pine

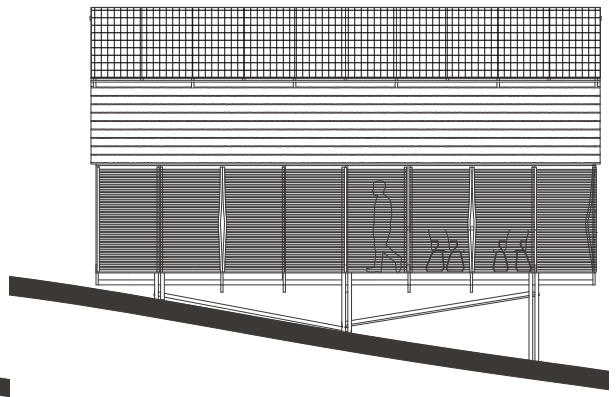
Floor

1. Flooring
2. Subfloor
3. Vapor barrier
4. Insolation +
Wood beams 295mm
5. Wood stits
6. Screw foundation



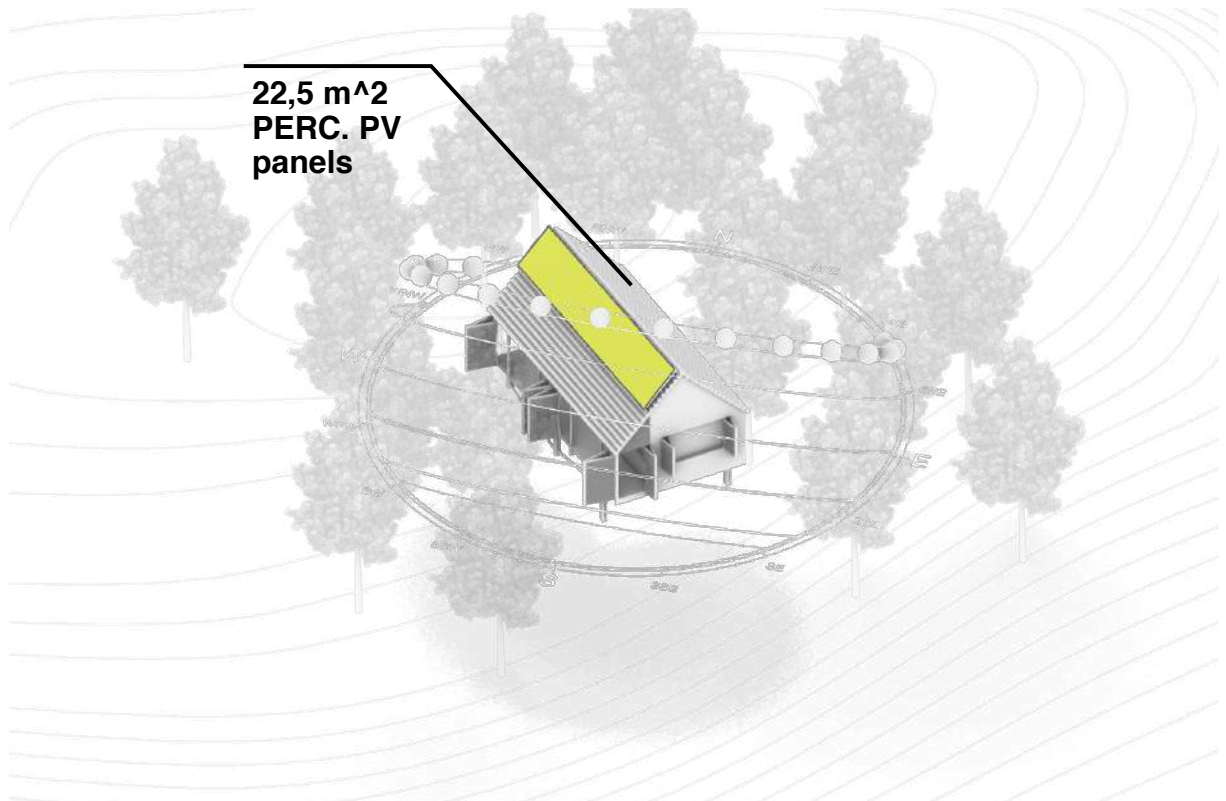


Cabin open



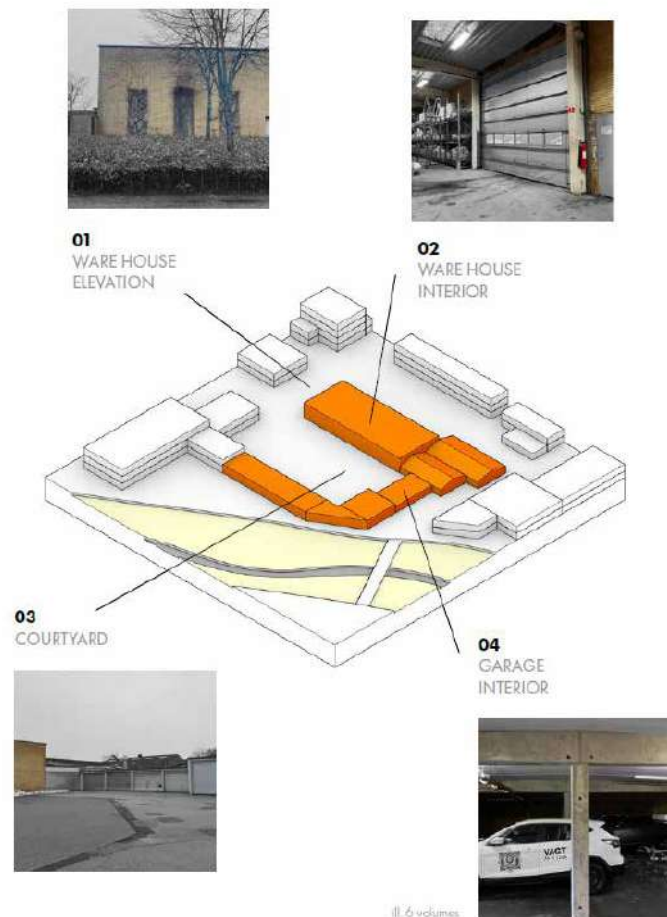
Cabin closed

The shutters were designed using Grasshopper and implemented to emphasize the cabin as a dynamic entity that can be opened or closed, responding to the relationship between the environment and the resident.



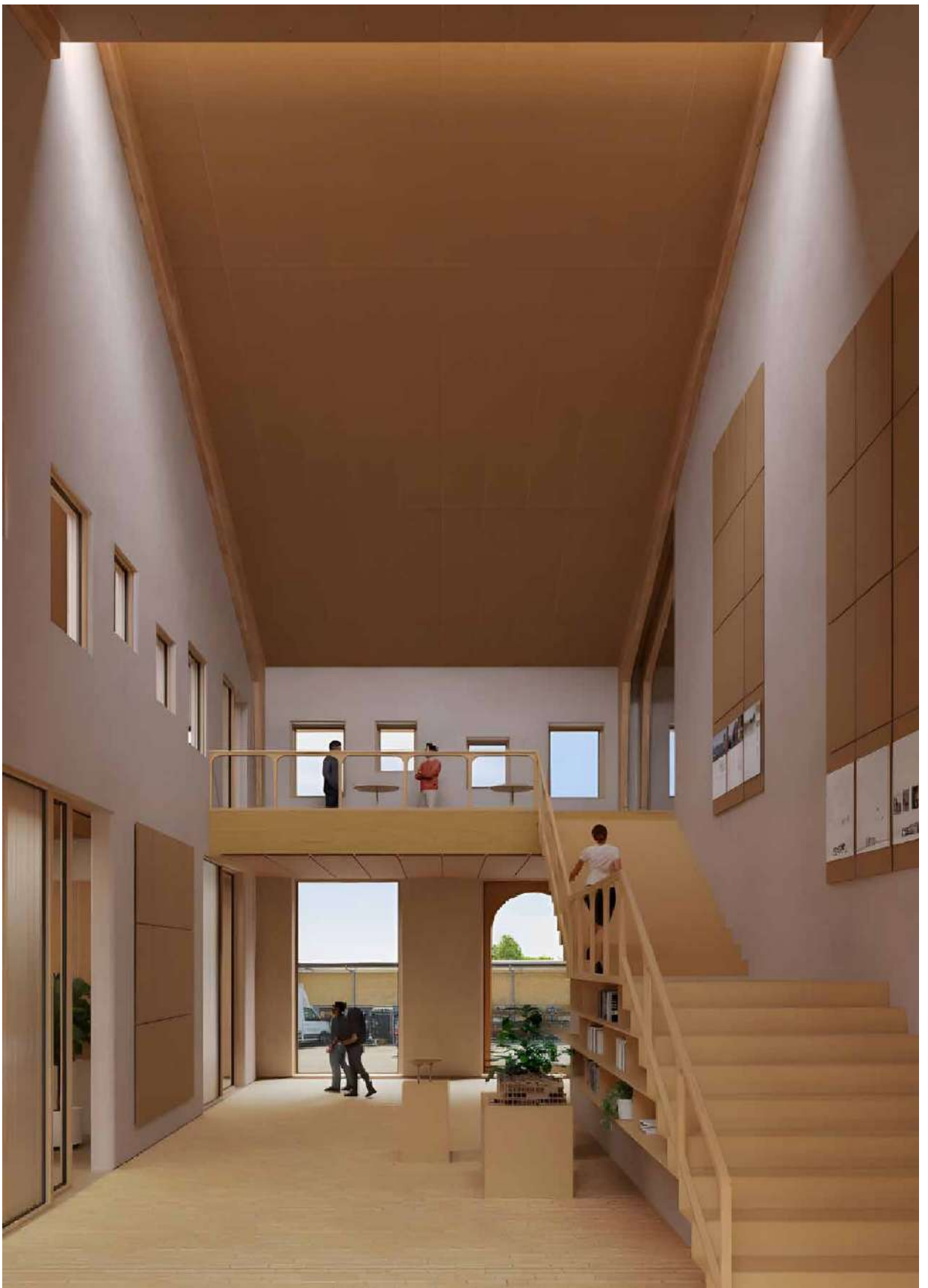
Transformation

Warehouse to School and community hub



Hjulumagerkvarteret is designated as a 'testbed' for this transition towards a circular economy. The area is planned to support a mix of residential, office, cultural, retail, and recreational spaces. This project's design proposal will become one of the first of these functions.

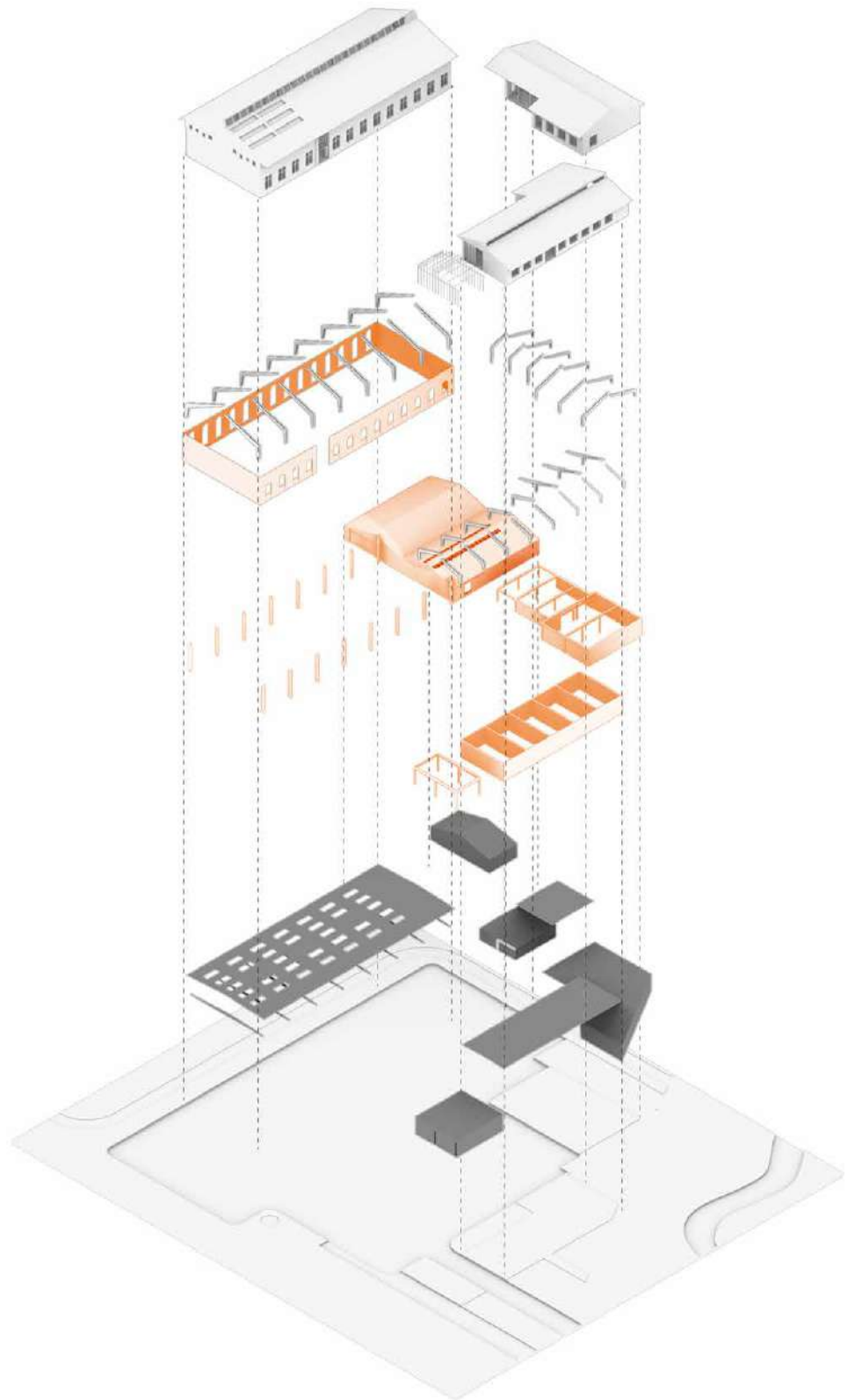
Problem statement: How can transformative and sustainable approaches, such as adaptive reuse, be utilised to repurpose a group of existing industrial structures into an architectural node facilitating collaborations between architectural professionals and stakeholders that aims to catalyse the development of the Hjulumagerkvarter area?

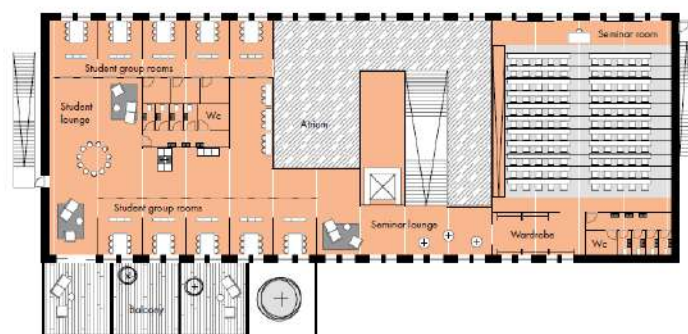
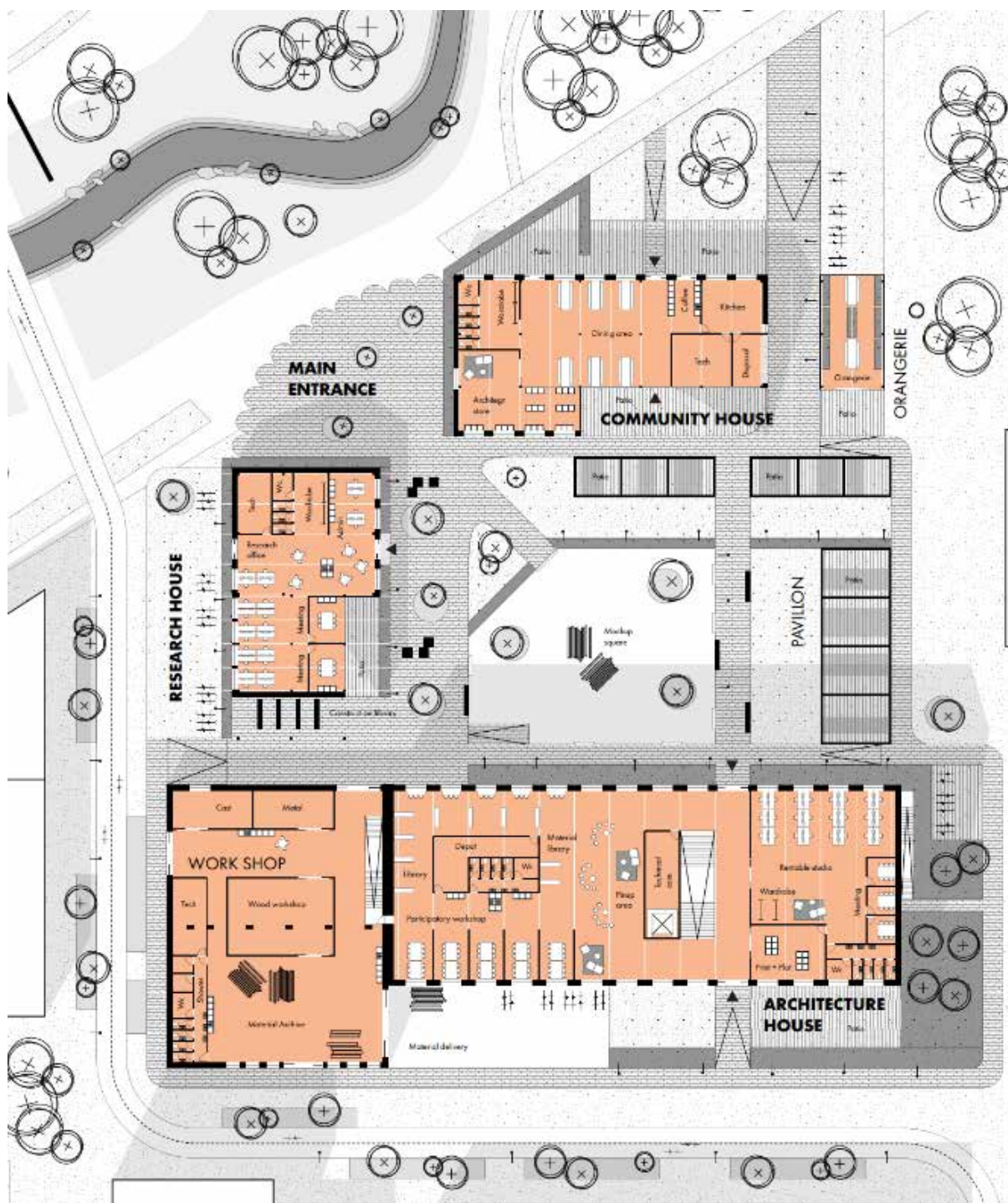


Added

Re-purposed

Removed





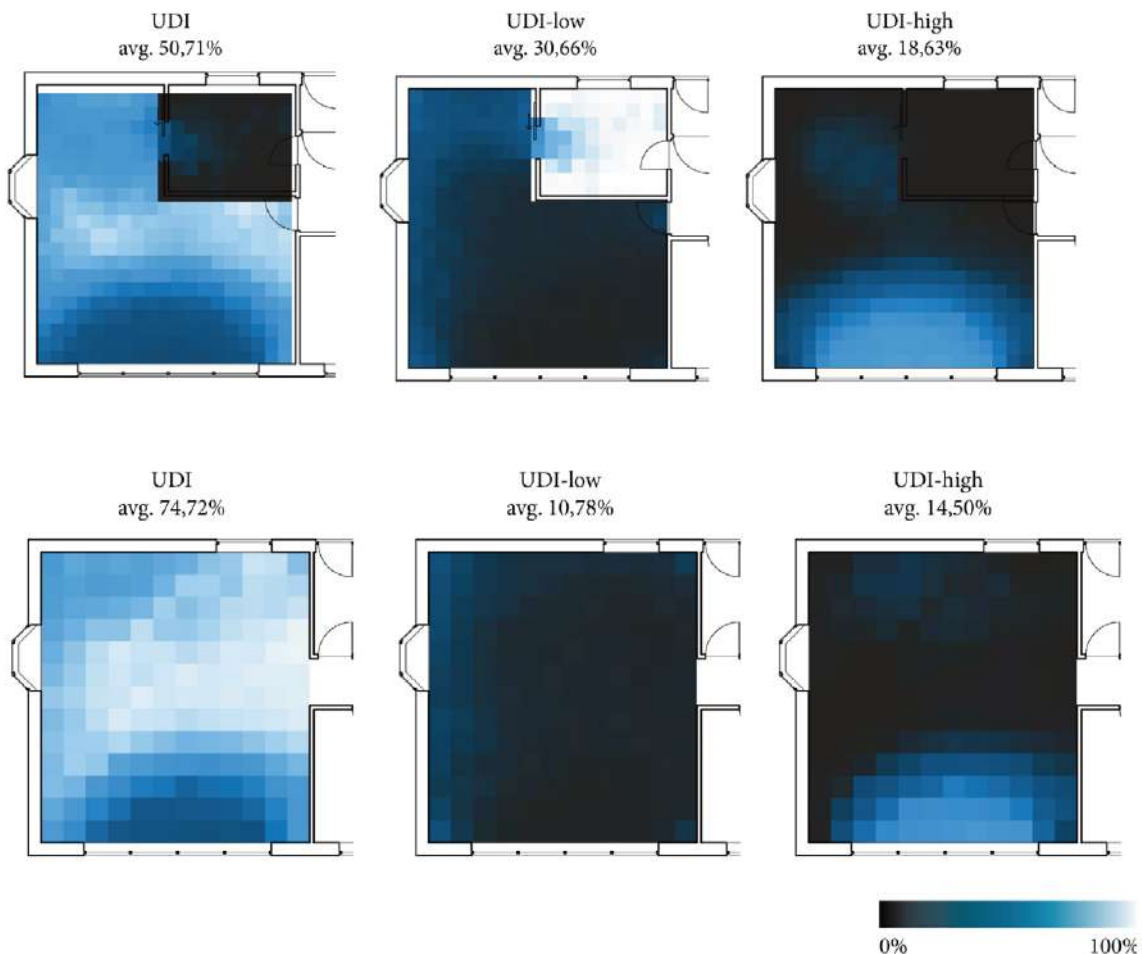
1. floor ill. 19 plan ARCHIVE

Extra

Indoor climate - Daylight and DGNB

During my internship at LINK Architektur, I worked with daylight simulations as part of the early design process. The analyses were used both to inform architectural decisions and to ensure compliance with BR18 daylight requirements. In addition, the results supported both Br18 and DGNB certification processes by documenting the projects' performance in relation to sustainability criteria.

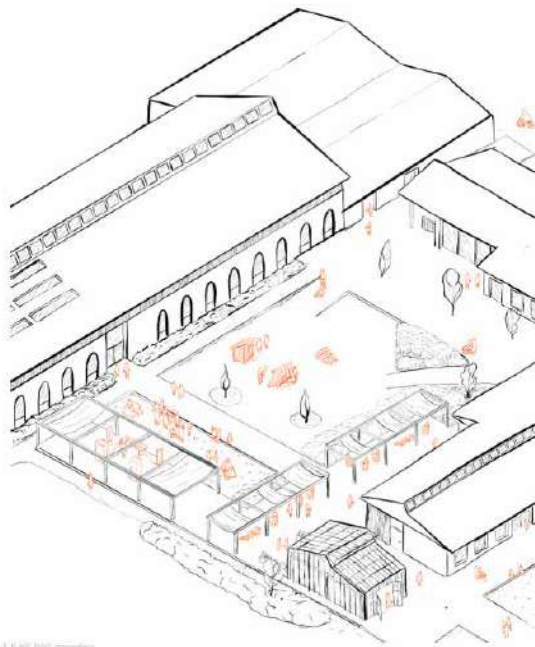
This experience gave me insight into how daylight studies can act as a design driver rather than just a regulatory check, bridging architectural quality, user comfort, and technical documentation.



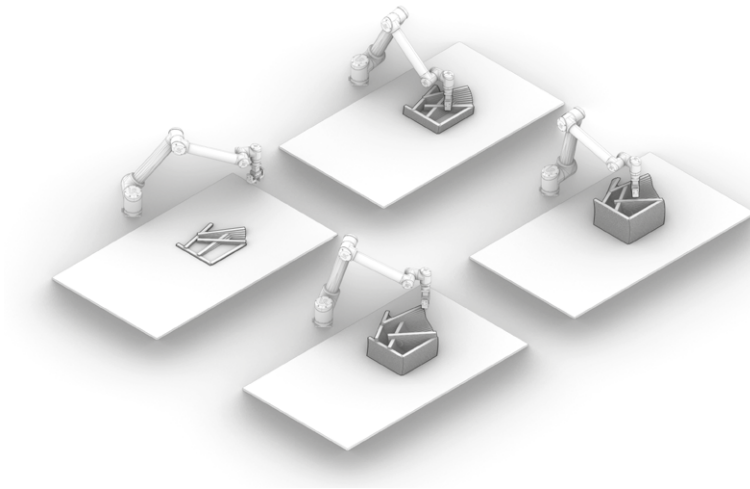
Life Cycle Assessment in Early and Detailed Design Phases

During my internship at LINK Arkitektur, I carried out Life Cycle Assessments (LCA) on projects at different stages, from early design concepts to detailed design phases. The analyses were used to guide material choices, evaluate environmental impact, and ensure compliance with BR18 requirements.

I gained experience with different tools and workflows, testing both in-house LCA software, online LCA platforms, and LCAByg. This work gave me hands-on insight into how LCA can be integrated into the design process, where environmental performance becomes an active parameter alongside functionality and aesthetics.

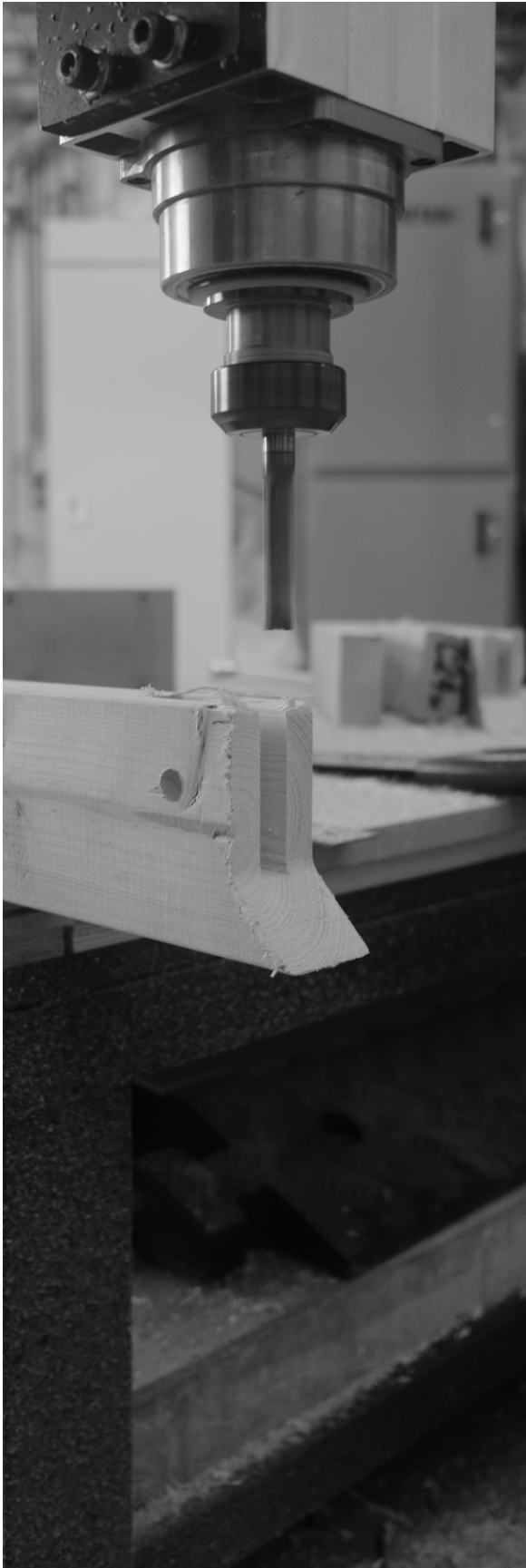


Robotics



Stacking, Illustration

Through two workshops, I explored robotic fabrication and robot-assisted human fabrication, gaining deeper insight into how automation can shape the future of construction. The experiments focused on material stacking, 5-axis milling, and the replication of Sashimono (traditional Japanese joinery). These experiences later informed my thesis *Designing with Scarcity*, in which I developed robotic milling toolpaths for fabricating structural elements from reclaimed timber.



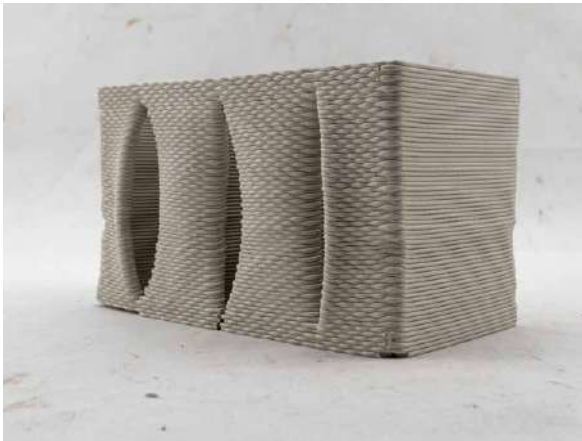
Milling Process, Photo



Stacking product, Photo



Clay print



I explored additive manufacturing through clay printing, focusing on the design of an acoustically optimized brick system. Each brick was parametrically modeled to respond to a specific acoustic scenario and printed to be stacked into a larger wall-like structure. Alongside this, I experimented with more expressive forms, including a reinterpretation of a vase by D. Diego Garcia Cuevas. Together, these studies deepened my understanding of how material behavior and print logic influence geometry and surface texture.



